

- Note:**
1. Question paper consists of two parts (**Part A and Part B**)
  2. Answer all sub questions from **Part A**
  3. Answer all questions from **Part B** with **either or choice**

| PART – A   |    |                                                                                                                                                                                         | 10 X 2 = 20 M |     |       |
|------------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----|-------|
|            |    |                                                                                                                                                                                         | CO            | BTL | Marks |
| I.         | a) | Distinguish between interference and diffraction.                                                                                                                                       | CO1           | L3  | 2 M   |
|            | b) | Discuss the concept of a quarter wave plate.                                                                                                                                            | CO1           | L6  | 2 M   |
|            | c) | Define the terms electric displacement vector and polarizability.                                                                                                                       | CO2           | L1  | 2 M   |
|            | d) | Obtain the relation between relative permeability and susceptibility.                                                                                                                   | CO2           | L2  | 2 M   |
|            | e) | What is primitive cell? Give examples                                                                                                                                                   | CO3           | L1  | 2 M   |
|            | f) | List the steps to determine the Miller indices of lattice planes in a crystal system.                                                                                                   | CO3           | L1  | 2 M   |
|            | g) | What are the limitations of wavefunction?                                                                                                                                               | CO4           | L2  | 2 M   |
|            | h) | Outline the merits of Classical Free electron theory of solids.                                                                                                                         | CO4           | L2  | 2 M   |
|            | i) | Write the expressions for density of electrons and holes in an intrinsic semiconductor.                                                                                                 | CO5           | L1  | 2 M   |
|            | j) | Mention the applications of Hall Effect.                                                                                                                                                | CO5           | L1  | 2 M   |
| PART – B   |    |                                                                                                                                                                                         | 5 X 10 = 50 M |     |       |
| UNIT - I   |    |                                                                                                                                                                                         |               |     |       |
| 1.         | a) | Explain the formation of colours in thin films.                                                                                                                                         | CO1           | L3  | 5 M   |
|            | b) | Obtain the conditions for central maxima and secondary minima , maxima in Fraunhofer diffraction due to single slits                                                                    | CO1           | L2  | 5 M   |
| (OR)       |    |                                                                                                                                                                                         |               |     |       |
| 2.         | a) | Describe the construction, principle and working of Nicol prism.                                                                                                                        | CO1           | L3  | 5 M   |
|            | b) | Newton’s rings are observed in reflected light of wavelength 5900 Å <sup>0</sup> . The diameter of the 10 <sup>th</sup> dark ring is 0.5 cm. Find the radius of curvature of lens used. | CO1           | L3  | 5 M   |
| UNIT – II  |    |                                                                                                                                                                                         |               |     |       |
| 3.         | a) | Derive an expression for electronic polarizability of dielectrics in terms of atomic radius.                                                                                            | CO2           | L2  | 5 M   |
|            | b) | Derive the expression for local field.                                                                                                                                                  | CO2           | L2  | 5 M   |
| (OR)       |    |                                                                                                                                                                                         |               |     |       |
| 4.         | a) | Explain hysteresis curve of ferromagnetic materials in detail.                                                                                                                          | CO2           | L3  | 5 M   |
|            | b) | Distinguish between soft and hard magnetic materials,                                                                                                                                   | CO2           | L4  | 5 M   |
| UNIT – III |    |                                                                                                                                                                                         |               |     |       |
| 5.         | a  | Obtain the packing factors of Simple cubic, Body centered cubic and face centered cubic structures.                                                                                     | CO3           | L3  | 10 M  |
| (OR)       |    |                                                                                                                                                                                         |               |     |       |
| 6.         | a) | Determine the interplanar distance of separation in a cubic crystal system.                                                                                                             | CO3           | L5  | 5 M   |
|            | b) | State and explain Bragg’s law.                                                                                                                                                          | CO3           | L2  | 5 M   |
| UNIT – IV  |    |                                                                                                                                                                                         |               |     |       |
| 7.         | a) | Explain de-Broglie’s concept of wave particle duality.                                                                                                                                  | CO4           | L3  | 5 M   |
|            | b) | Analyse the behaviour of a free particle in one dimensional potential well and show that energy of the particle confined to the well is quantized.                                      | CO4           | L3  | 5 M   |
| (OR)       |    |                                                                                                                                                                                         |               |     |       |

|                 |           |                                                                                                        |            |           |            |
|-----------------|-----------|--------------------------------------------------------------------------------------------------------|------------|-----------|------------|
| <b>8.</b>       | <b>a)</b> | Explain Fermi Dirac distribution and its temperature dependance, Illustrate it graphically.            | <b>CO4</b> | <b>L4</b> | <b>5 M</b> |
|                 | <b>b)</b> | Derive an expression for electrical conductivity using quantum free electron theory.                   | <b>CO4</b> | <b>L5</b> | <b>5 M</b> |
| <b>UNIT – V</b> |           |                                                                                                        |            |           |            |
| <b>9.</b>       | <b>a)</b> | Write a short note on extrinsic semiconductors.                                                        | <b>CO5</b> | <b>L2</b> | <b>5 M</b> |
|                 | <b>b)</b> | Analyse the dependance of Fermi level on concentration of charge carriers in extrinsic semiconductors. | <b>CO5</b> | <b>L4</b> | <b>5 M</b> |
| <b>(OR)</b>     |           |                                                                                                        |            |           |            |
| <b>10.</b>      | <b>a)</b> | What do you understand by drift and diffusion currents in semiconductor? Derive their expressions.     | <b>CO5</b> | <b>L2</b> | <b>5 M</b> |
|                 | <b>b)</b> | Deduce Einstein's equations for semiconductors.                                                        | <b>CO5</b> | <b>L5</b> | <b>5 M</b> |

**\*\*\* End of the Question Paper \*\*\***